

A knowledge-intensive LLM-assisted evolutionary framework for multi-objective geometric design of high-performance concrete structures

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ABSTRACT

The design of high-performance building components, such as hollow concrete blocks with intricate internal geometries, presents significant challenges in exploring vast search space and balancing conflicting objectives. Traditional evolutionary algorithms (EAs), while effective in global search, require substantial human effort and domain knowledge to tune genetic operators when addressing such complex, knowledge-intensive tasks. To overcome these limitations, this study proposes a novel framework that integrates a Large Language Model (LLM) with the Non-dominated Sorting Genetic Algorithm II (NSGA-II) for multi-objective geometric optimization. Within this framework, the LLM functions as an intelligent solution generator, periodically injecting diverse candidates to nudge the evolutionary process toward superior Pareto fronts. To enable efficient evaluation of complex designs, the framework also incorporates pre-trained Graph Neural Network (GNN) surrogate models for rapid performance prediction. A systematic investigation of two key LLM-related parameters, intervention frequency (F) and the number of generated solutions (N), revealed that the LLM-assisted approach outperforms standard NSGA-II. The best configuration (F = 10, N = 10) achieves a 30.9% reduction in Inverted Generational Distance (IGD) and a 7.3% increase in Hypervolume (HV), demonstrating substantial gains in both convergence and diversity of the Pareto front. This study demonstrates the potential of LLMs in supporting knowledge-intensive, geometry-driven optimization tasks and offers a user-friendly paradigm for high-performance structural design.

1. Introduction

Modern engineering disciplines are increasingly confronted with complex design challenges that demand innovative solutions for optimized performance, resource efficiency, and sustainability. Within the building sector, these demands for multi-functional and high-performance building components are particularly escalating (Ashraf and Abidin, 2024). A key driver of this trend is the global movement toward decarbonization, which calls for more sustainable construction practices and materials to reduce environmental impact (Räihä et al., 2024; Yang et al., 2023). Concurrently, there is also a growing demand to maintain or even enhance comfort levels within built environments, as people in modern society spend almost 90% of their time indoors (Liu et al., 2024). To simultaneously address these environmental and occupant-centered goals, the development of advanced building components, such as hollow concrete blocks, plays a pivotal role (Sathiparan

et al., 2014). These components offer significant benefits in terms of lightweight construction, reduced material consumption (Çavuş et al., 2025), improved structural reliability (Lin et al., 2025; Dang et al., 2019), and enhanced human comfort (Wang et al., 2025; Wang et al., 2023). A key factor that influences these characteristics is the shape of voids or hollow spaces in these components (Al-Tamimi et al., 2020). However, optimizing their intricate internal geometries to simultaneously achieve superior properties in different aspects, especially those based on structures like irregular polygonal air cavities (Wang et al., 2024), represents a complex design endeavor. This process necessitates sophisticated design exploration, multi-objective trade-off analysis, and a deep understanding of material behavior, thereby underscoring the critical role of advanced computational methods in supporting such knowledge-intensive engineering tasks (Wang et al., 2025).

Traditionally, the design optimization of such components heavily relies on high-fidelity numerical simulations, such as Finite Element

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Analysis (FEA), to evaluate the performance of each candidate (Li et al., 2025). While accurate, FEA is computationally intensive for iterative design exploration across vast design spaces, limiting the efficiency of optimization efforts (Awal et al., 2025). Conventional metaheuristic algorithms, including evolutionary algorithms (EAs) like the Non-dominated Sorting Genetic Algorithm II (NSGA-II) (Deb et al., 2002), are powerful tools for navigating these large search spaces (Wang et al., 2026). However, their effective application often demands substantial human effort and domain knowledge for the meticulous adjustment of genetic operators (Liu et al., 2024; Lee et al., 2024). Without such careful tuning, these algorithms are prone to issues like premature convergence or stagnation, hindering their ability to thoroughly explore the design space and discover truly optimal solutions (Li et al., 2024; Gutiérrez-Aguirre and Contreras-Bolton, 2024; Xia et al., 2026). These limitations highlight a pressing need for more intelligent and efficient approaches to accelerate and enhance the discovery of optimal engineering designs.

In recent years, the rapid advancements in Artificial Intelligence (AI), particularly Large Language Models (LLMs), have revolutionized various fields by demonstrating remarkable capabilities in understanding, generating, and reasoning with human-like text (Su et al., 2026). Trained on vast amounts of text data to perform next-token prediction (Cheng et al., 2024), LLMs possess an implicit understanding of complex patterns and relationships, enabling them to extract knowledge and even generate conceptually novel outputs (Jiang et al., 2025). This inherent ability to synthesize information and propose creative solutions presents an unprecedented opportunity to augment traditional engineering design and optimization processes. By integrating LLMs, it becomes possible to inject an element of reasoning intelligence into the search process, potentially guiding the exploration more effectively than purely random or heuristic operations.

Addressing the aforementioned challenges and leveraging the emerging capabilities of LLMs, this study proposes a novel LLM-assisted evolutionary framework for the multi-objective optimization of hollow concrete components with irregular polygonal air cavities, targeting enhanced thermal and mechanical performance. Our primary contribution lies in demonstrating how LLMs can be strategically integrated

into a genetic algorithm (GA) to overcome the limitations of traditional stochastic search. Specifically, the LLM periodically generates new candidate solutions, which are then judiciously introduced into the population to replace sub-optimal individuals, thereby fostering a more efficient and effective exploration of the design space. At each LLM intervention event, the geometric constraints, objective specifications, and current elite solutions are encoded into a structured natural-language prompt, constituting a structured knowledge representation that dynamically conveys domain expertise to the LLM. This design draws on the established paradigm of knowledge-intensive expert systems, in which domain knowledge is explicitly represented and made available to a reasoning component (Gindullina et al., 2026). This research uniquely focuses on complex geometric design optimization, a domain where the efficacy of LLMs in handling spatial relationships and generating viable structural configurations remains largely unexplored.

Furthermore, to mitigate the computational burden of high-fidelity simulations, pre-trained Graph Neural Network (GNN) (Wang et al., 2025) surrogate models are employed, enabling rapid and accurate performance evaluation of candidate designs (Liu et al., 2026; Li et al., 2026). Beyond demonstrating enhanced optimization performance, this study also investigates the impact of key LLM intervention parameters, namely the frequency of LLM intervention and the number of LLM-generated solutions. Through comparative analysis, we aim to provide practical insights and actionable recommendations for effectively deploying LLM-assisted optimization in complex engineering tasks.

The research gap addressed in this study was identified through a literature screening of peer-reviewed research papers conducted using the Web of Science database, as illustrated in Fig. 1. An initial broad query with the keyword “large language model” returned a substantial number of documents, which indicates a burgeoning interest in this field. This was subsequently refined to identify studies at the intersection of LLMs and specific optimization methodologies or scopes. The refined searches yielded limited results: “large language model & genetic algorithm” returned 60 documents, “large language model & multi-objective optimization” returned 50 documents, and “large language model & shape optimization” returned 29 documents. Importantly, a final, more focused search combining all four keywords yielded

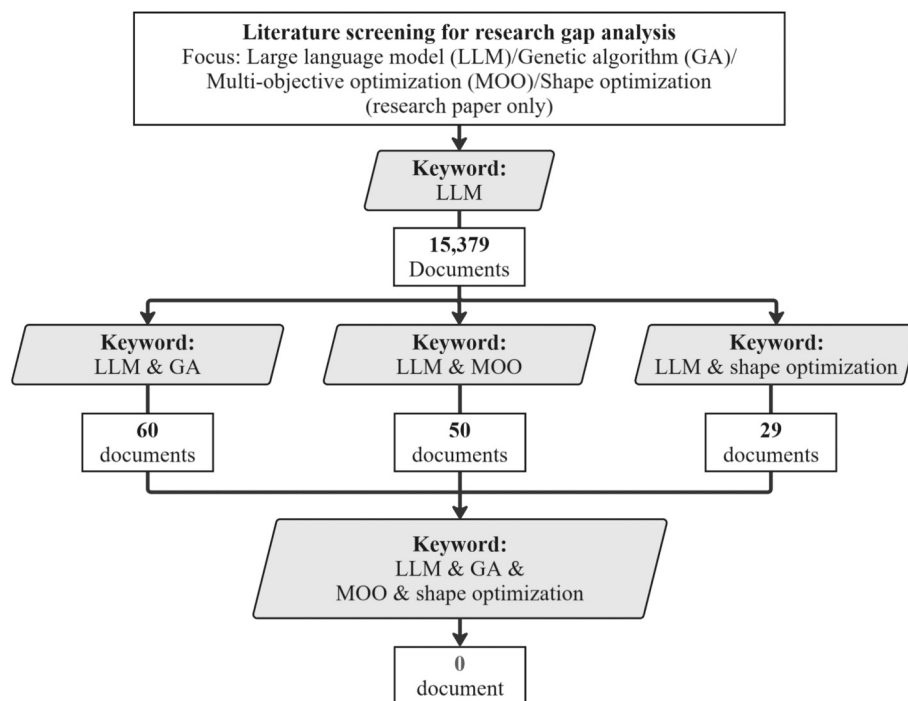


Fig. 1. Literature screening workflow for identifying research gaps in large language model-assisted genetic algorithm in multi-objective shape optimization. Last assessed on February 18, 2026.

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This absence highlights a specific and largely unexplored research opportunity at the intersection of AI and structural engineering. While individual components of this research have been explored, no published work was found that concurrently investigates all three functionalities in a single framework. To the best of the authors' knowledge, this study pioneers the integration of LLMs into a surrogate-assisted evolutionary algorithm (SAEA) (Wei et al., 2023) specifically tailored for knowledge-intensive geometric design in the building and construction domain. By addressing intricate spatial relationships and multi-objective trade-offs, this work establishes a new paradigm for accelerating the discovery of high-performance designs in engineering fields, offering a more intuitive control over the optimization process through natural language instructions.

The proposed framework can be interpreted through the lens of knowledge-based systems, a core theme in expert systems research. Within this paradigm, the LLM serves as an implicit expert knowledge base, leveraging domain knowledge encoded during pre-training on extensive scientific and engineering literature. The prompt engineering mechanism constitutes a structured form of knowledge representation, through which structural design objectives, geometric constraints, and performance trade-offs are explicitly communicated to guide the LLM's reasoning. The LLM's process of analyzing patterns in elite solutions and generating novel candidate designs constitutes a form of knowledge-driven reasoning, analogous to how a domain expert would synthesize known high-performing precedents to propose new design directions. This knowledge-intensive paradigm distinguishes the proposed framework from conventional AI-assisted optimization approaches and positions it within the broader scope of intelligent and expert systems research.

The remainder of this paper is organized as follows. Section 2 summarizes the related work in terms of the use of LLM in evolutionary algorithms in existing studies. Section 3 outlines the overall workflow, including parametric modeling for hollow concrete components, the development of the surrogate models, and the experimental setup for the integration of the LLM and NSGA-II. Section 4 presents the experimental results, beginning with an evaluation of the solution quality between NSGA-II and the LLM-assisted framework, followed by a comparative analysis of impact of LLM intervention, and together with a discussion of the guidance in practical applications. Section 5 provides concluding remarks, discusses limitations, and outlines future research directions.

2. Related works

The integration of LLMs and EAs represents a synergistic paradigm by leveraging the distinct strengths of each methodology to address complex optimization challenges (Wu et al., 2024). Evolutionary algorithms, exemplified by the NSGA-II for multi-objective optimization, are renowned for their robust capabilities in exploring vast and intricate design spaces to identify optimal or near-optimal solutions (Liu et al., 2025). Their strength in population-based search and handling multi-objective trade-offs are well-documented in existing literature (Chen et al., 2026; Weng et al., 2026). Conversely, LLMs, trained on extensive textual datasets, possess remarkable abilities in knowledge extraction, pattern recognition, and reasoning (Liu et al., 2025). This enables them to interpret complex problem descriptions, infer underlying relationships, and generate novel solutions that might be informed by human expertise or design principles implicitly learned from their training data (Mustapha, 2025). Therefore, the integration between LLM and EA is not a mere combination but a strategic fusion where LLMs can enrich the evolutionary process by introducing intelligent solutions (Wu et al., 2024).

Recent studies have begun to explore the integration of LLMs into traditional EAs through various strategic approaches. For instance, Reinhart and Statt (Reinhart and Statt, 2024) applied an LLM to the design of sequence-defined macromolecules via evolutionary

optimization, and found that the LLM-based optimizer often converged more rapidly to high-quality solutions compared to conventional methods. Similarly, Liu et al. (Liu et al., 2024) proposed a multi-objective evolutionary optimization framework in which the LLM functions as a solution generator, and demonstrated that their framework outperformed existing EAs that rely on handcrafted search operators. Beyond solution generation, some researchers have investigated the feasibility of using LLMs to replace traditional genetic operators. For example, Meyerson et al. (Meyerson et al., 2024) showed that LLMs can produce heritable variations when used as a substitute for crossover operations within an evolutionary process. In another study, Brahmachary et al. (Brahmachary et al., 2024) employed LLMs to replace both crossover and mutation in a standard genetic algorithm, and found that the LLM exhibited clear advantages in solving low-dimensional problems compared to traditional approaches. Moreover, LLMs have been leveraged to enhance other components of evolutionary search, such as generating promising initial populations (Huang et al., 2025), selecting parents from population (Ali et al., 2025), or serving as evaluators to assess the quality and diversity of candidate solutions (Bradley et al., 2023). Collectively, these preliminary investigations provide encouraging evidence of the potential for LLMs to enhance both the efficiency and effectiveness of EAs.

However, a significant gap remains in the current literature. Most existing applications of LLM-assisted evolutionary optimization are predominantly confined to problems that can be characterized as low-dimensional or semantically transparent, that is, problems in which the solution representation consists of simple numerical vectors or symbolic sequences with well-defined mathematical or combinatorial structure, such as pure mathematical benchmark functions (Meyerson et al., 2024), combinatorial tasks including the knapsack problem (Huang et al., 2025) and the traveling salesman problem (Ali et al., 2025), text-based design tasks (Yang and Li, 2023), or mathematically formulated engineering problems where the relationship between design variables and objectives can be expressed through closed-form equations (Brahmachary et al., 2024; Zhong et al., 2025). In these settings, the LLM can directly interpret the meaning of each variable and reason over the relationship between solution components and performance outcomes, because the mapping from design encoding to physical behavior is explicit and algebraically transparent.

The problem addressed in this study, by contrast, belongs to a fundamentally different class: spatially complex, geometrically implicit optimization. The design variables (e.g. the case in this study) do not encode performance directly; their physical meaning emerges only through a non-linear and topology-dependent process of spatial construction. The arrangement of seeds determines the Voronoi cell structure, which in turn governs heat flow paths and mechanical load transfer through the resulting concrete frame. There is no closed-form relationship between the input coordinates and the output objectives; the mapping is both high-dimensional and geometrically nonlinear. This class of problems involves the computational or cognitive capacity to deduce topological relationships and physical consequences from the relative positioning of entities within a Euclidean coordinate system, and poses a distinct challenge for traditional evolutionary methods.

EAs are powerful precisely because they are general-purpose solvers capable of operating on any problem that can be encoded as a numerical string (Wu et al., 2024). However, this generality comes at a cost: their genetic operators function purely on the basis of mathematical and probabilistic rules, without any awareness of the physical or geometric meaning embedded in the encoded designs. When the crossover operator recombines two parent solutions, it operates on numerical values rather than spatial configurations, and can readily produce offspring in which geometrically incompatible features are combined, destroying the high-performing structural patterns present in both parents (Vijai, 2025). Similarly, the mutation operator randomly perturbs numerical values without any comprehension of the geometric consequences, which means that while it is intended to diversify the population and

escape local optima, it can equally disrupt beneficial spatial arrangements and lead to performance degradation (Wang et al., 2025). In short, because EAs operate without understanding the semantic content of the design representation, each crossover or mutation functions as an uncertain trial with no guarantee that the resulting change aligns with geometric design intent or improves structural performance. This semantic blindness is a manageable limitation for problems with explicit mathematical structure, where performance gradients are predictable, but becomes a substantive obstacle in spatially complex domains where the relationship between design variables and performance is implicit and topology-dependent.

It is precisely this gap that motivates the integration of LLMs into the evolutionary search process. Unlike GA operators, which manipulate design encodings without geometric awareness, LLMs possess the capacity to reason over the spatial semantics of the encoded configurations through their pre-trained understanding of geometric relationships and design patterns. This reasoning capability has the potential to complement the general-purpose search of EAs by introducing candidates that reflect structured geometric logic rather than random numerical perturbation. As such, the potential of LLMs to address optimization problems involving complex geometric reasoning, particularly in multi-objective contexts, remains largely unexplored. Notably, there is an absence of research applying LLM-assisted EAs to shape optimization, which is a central concern in many architectural and engineering design tasks. This represents a missed opportunity, especially for the building and construction domain, where geometry plays a crucial role in both performance and feasibility (Al-Tamimi et al., 2020).

In the domain of Architecture, Engineering, and Construction (AEC), LLM-assisted design is still in its infancy (Kampelopoulos et al., 2025). Current applications of LLMs in AEC are primarily concentrated in the following areas: (i) knowledge support (e.g., education (Cruz Castro et al., 2024) and bibliometric analysis (Meng et al., 2024)), (ii) project coordination and compliance (e.g., construction scheduling (Prieto et al., 2023) and task planning (You et al., 2023)), (iii) intelligent assistance (e.g., virtual assistants (Lu et al., 2024) and question-answering systems (Zhong and Goodfellow, 2024)), and (iv) model and data interpretation (e.g., BIM functionalities (Jang et al., 2024), as well as the associated information visualization (Zheng and Fischer, 2023)). While these applications demonstrate the utility of LLMs in improving efficiency, they do not directly engage with the core challenge of geometry-driven design optimization in the field of AEC. To date, no existing studies have applied LLM-assisted EAs to the optimization of concrete structures, despite the critical role of shape in their performance. This absence again highlights a significant research gap for the adoption of LLM-based methods in building-related optimization problems.

Hence, this study proposes a novel framework specifically designed for the multi-objective shape optimization of building components, which are characterized by complex geometric configurations. By focusing on this spatially dependent problem, the research seeks to evaluate the potential of LLMs to generate structurally viable and high-performing candidate solutions. In addition, the framework incorporates pre-trained surrogate models to accelerate performance evaluation, offering a new hybrid paradigm that combines the generative capabilities of LLMs with the efficiency of surrogate-assisted evolutionary search. This approach not only fills a critical gap in the literature but also contributes to a practical methodology for advancing AI-supported design in the built environment.

Another critical limitation in existing research is the insufficient focus on systematically analyzing the impact of various LLM-related parameters on the overall optimization performance. Current studies often concentrate on a singular integration strategy between LLM and EA without a deeper analysis of factors such as LLM intervention frequency and the quantity of LLM-generated solutions. The lack of standardized experimental setups, coupled with inconsistencies in optimization cases, LLM configurations, and injection strategies across

different studies, makes direct comparison and generalization of findings exceptionally challenging. Consequently, this fragmented landscape impedes our understanding of how to effectively harness LLMs within evolutionary frameworks and hinders their broader adoption in engineering practice. To address these issues, this study conducts a controlled investigation into different LLM intervention strategies, systematically varying the frequency of LLM activation and the quantity of LLM-generated solutions. By embedding these variations within a consistent optimization scenario, this experimental design ensures comparability and enables robust insights into their effects. The findings not only inform the design of more effective LLM-assisted frameworks for optimizing complex concrete structures, but also offer practical guidance for other domains involving intricate geometric configurations.

3. Methodology

This section begins with an overview of the entire framework of this study, followed by detailed information about each step. The methodology is structured into a three-step framework, as illustrated in Fig. 2. The pipeline starts with (i) *Parametric modeling*, where the design space of irregular hollow concrete components is defined and their physical properties are evaluated via numerical simulation. Due to the high computational cost of FEA, the next step involves (ii) *Surrogate model construction*, where two GNNs are trained on a large dataset to rapidly predict material properties from geometric inputs. The final and core stage of this work is (iii) *LLM-assisted optimization*, where a novel framework integrating an LLM with the NSGA-II algorithm is developed to efficiently search the design space for optimal solutions. Detailed information on each step will be presented in the following sections.

3.1. Parametric modeling

In this paper, we focus on the shape optimization of customized concrete components, which can be fabricated with 3D-print-enabled approach (Wu et al., 2021; Lu et al., 2025) and are significant in the decarbonization efforts within architectural engineering (Wang et al., 2026). The methodology begins with the parametric definition of a block prototype from a concrete partition wall, as detailed in Fig. 3. The prototype is a $50 \times 50 \times 50$ mm cube with a complex internal void structure, following the design principles and dimensions from Refs. (Wang et al., 2023; Wang et al., 2024). This structure is generated using a 2D Voronoi tessellation (Pokojski and Pokojaska, 2018), where the coordinates (x,y) of 5 seed points serve as the primary design variables. Controlled by a set of seed points, Voronoi diagrams partition space into efficient, non-overlapping regions (Long et al., 2024), which enables spatial distributions that meet certain needs for operators and designers across various fields (Yang et al., 2024), especially in designing tunable porous and cellular structures and materials (Quesada-Molina et al., 2025). The resulting 2D Voronoi pattern is extruded through the cube's height, and the entire geometry is set to maintain a constant porosity of 45% across all designs. The design of block prototype with irregular polygonal air cavities in this study is implemented via the parametric design software Rhino & Grasshopper (McNeel, 2020). Once a parametric model is generated, its performance is evaluated against two competing physical objectives using FEA.

3.1.1. Objective 1: Thermal conductivity

To assess the thermal insulation properties, a steady-state heat transfer analysis was performed. The analysis assumes that the heat transfer mechanism within the hollow concrete block is conduction-dominated, a common and validated assumption for such structures (Wang et al., 2023). The boundary conditions for the simulation were established based on previous literature (Wang et al., 2023) to replicate a typical indoor-outdoor temperature difference. The left boundary, representing the outdoor face, was maintained at a constant temperature

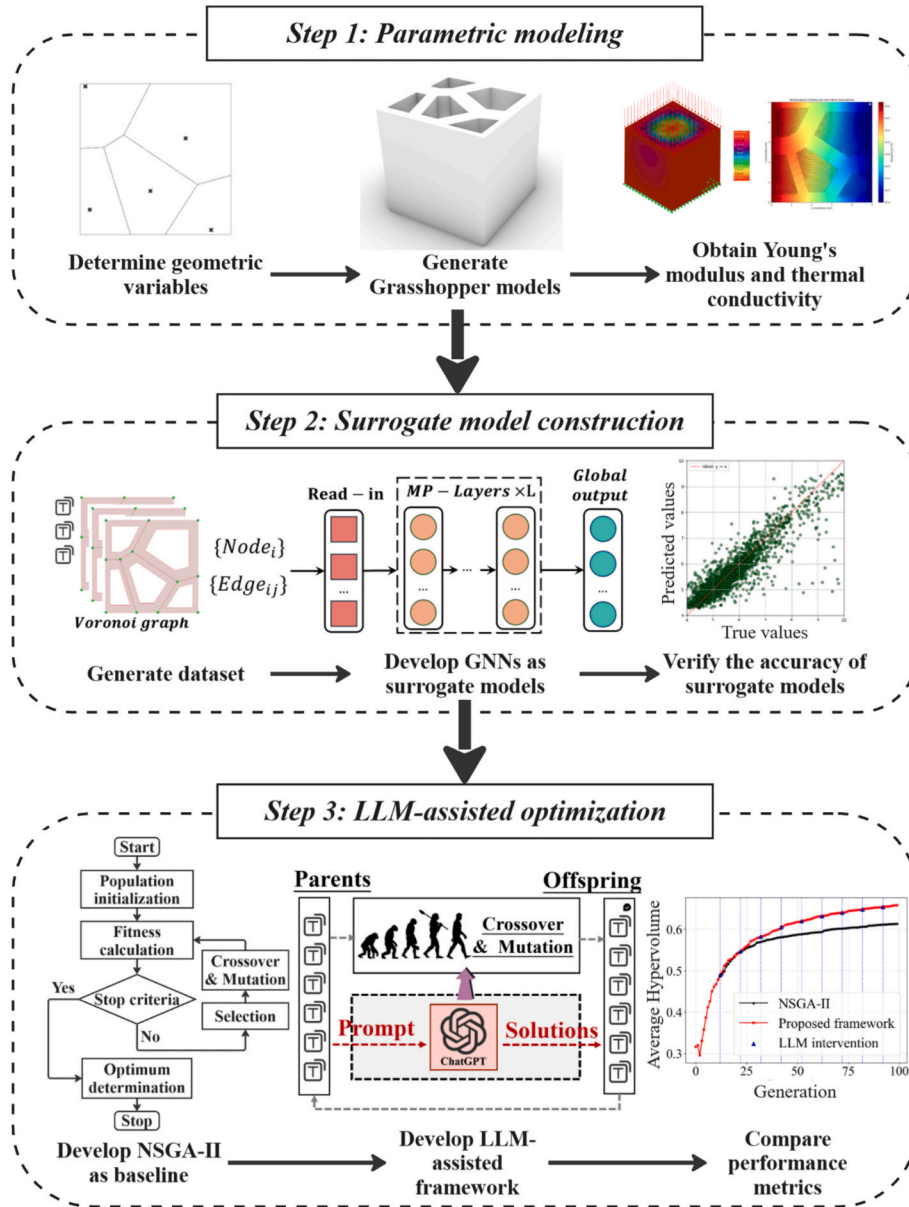


Fig. 2. The overall three-step workflow of the proposed research, encompassing parametric modeling, surrogate model construction, and LLM-assisted optimization.

of 35°C, while the right indoor face was set to 25°C. Other settings can be found in Table 1.

After solving for the temperature field across the prototype, the total heat flux can be calculated. The relationship between heat flux and the temperature gradient is governed by Fourier's Law:

$$q = K \frac{\Delta T}{\Delta x}$$

where q is the heat flux (W/m^2), K is the thermal conductivity ($W/(mK)$), ΔT is the temperature gradient cross the thickness Δx .

3.1.2. Objective 2: Young's modulus

To evaluate the mechanical integrity, a static structural analysis simulating a uniaxial compression test is performed. The boundary and loading conditions are established as follows, in line with Ref. (Wang et al., 2025): The bottom surface of the block is defined with a fixed support, with all nodes on this surface being fully constrained against translation in all dimensions and against rotation about all axes. A uniform compressive load is applied to the top surface by subjecting it to

a nodal displacement of 0.5 mm. The solid concrete portion of the model is defined with a base Young's modulus of 6400 MPa. The focus of the analysis is on the nodal reaction forces on the top surface, which are calculated by the Karamba3D solver. The Young's modulus is then computed by summing these nodal reaction forces to find the total force, calculating the average stress on the cross-section, and relating it to the applied strain (Wang et al., 2025).

3.2. Surrogate model construction

In contrast to conventional optimization algorithms that rely on predefined or explicit functional relationships (Zhong et al., 2024), the intricate connection between geometry appearance and the resulting thermal and mechanical properties of the structures lacks a straightforward analytical formulation. Therefore, two GNN-based surrogate models are developed as alternatives to the fixed function relationship. Compared with traditional FEA simulations requiring extensive repetitive computations (Li et al., 2025), the introduction of surrogate models can significantly accelerate performance prediction while preserving

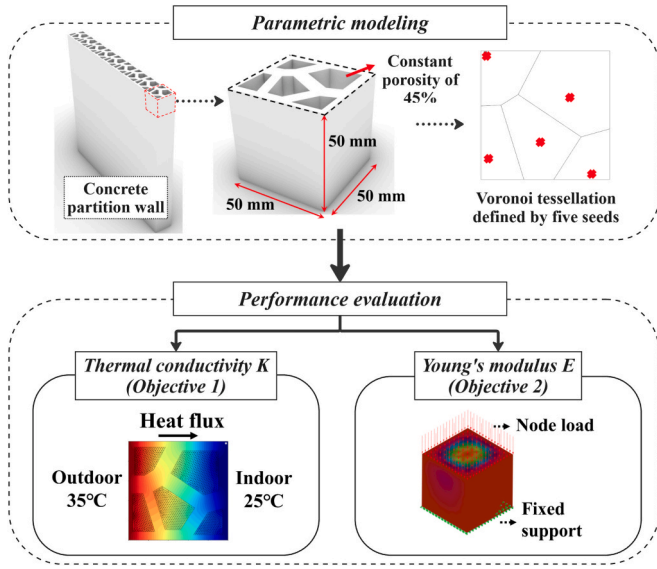


Fig. 3. The workflow for parametric modeling and performance evaluation.

Table 1
Simulation set-up for heat transfer.

Features	Value
Left boundary temperature	35°C
Right boundary temperature	25°C
Top and bottom	Adiabatic: heat flux 0 W/m ²
Conductivity (Concrete)	0.868 W/(m·K)
Conductivity (Air, insulation)	0.046 W/(m·K)
Convection coefficient (Indoor)	8.7 W/(m ² ·K)
Convection coefficient (Outdoor)	23.0 W/(m ² ·K)

fidelity (Jasiński et al., 2024; He et al., 2025).

To facilitate the training of these GNN models, a dataset was initially created. This involved conducting numerical simulations on 12,000 irregular polygonal structures to obtain their corresponding geometric information and the values for the two optimization objectives. Subsequently, this dataset was systematically partitioned for surrogate model development, allocating 80% for training and the remaining 20% for testing. The predictive performance of these two GNN models, as detailed in Table 2, demonstrates their sufficient accuracy. The accuracy, coupled with their computational efficiency, enables a rapid and precise evaluation of candidate designs for the subsequent LLM-assisted optimization phase.

3.3. The LLM-assisted framework for optimization

This section presents the detailed settings of the framework integrating LLM and NSGA-II. The proposed framework synergizes the robust search capabilities of NSGA-II with the abstract reasoning and pattern recognition abilities of the LLM. The core idea is to employ LLM as an intelligent assistant that generates candidate solutions to nudge the evolutionary process more efficiently.

Table 2
Performance metrics of the trained surrogate models on the test set.

Models	MSE	MAE	MRE
Surrogate model 1 (Thermal conductivity)	8.00E-05	7.32E-03	1.26E-02
Surrogate model 2 (Young's modulus)	2.25E-01	3.31E-01	4.80E-02

3.3.1. Baseline algorithm

Our framework builds upon the NSGA-II (Deb et al., 2002), an efficient meta-heuristic multi-objective genetic algorithm that is commonly deployed to solve multi-objective optimization problems in engineering applications. The basic workflow of NSGA-II involves the following key steps: (i) Initialization of a random population; (ii) Evaluation of objective functions for each individual; (iii) Generation of offspring through genetic operators: selection, crossover, and mutation; (iv) Combination of parent and offspring populations followed by non-dominated sorting and crowding distance sorting to retain the best individuals for the next generation (Deb et al., 2002). This process repeats until a predefined termination criterion (e.g., number of generations) is met. Note that the result of multi-objective optimization is not a single optimal solution, but rather a Pareto front comprising multiple trade-off solutions across objectives. In this study, the standard NSGA-II serves as the baseline for evaluating the effectiveness of the LLM-assisted approach.

In our case, each solution is represented as a 10-dimensional decision vector corresponding to five pairs of (x, y) seed coordinates that generate a Voronoi diagram. The algorithm initializes a population of such solutions and iteratively evolves through genetic operators. The multi-objective optimization problem can be defined as:

$$\begin{cases} \min[K(x, y)] \\ \min[-E(x, y)] \end{cases}$$

where x and y represent the variable of geometry appearance, K denotes the function of thermal conductivity regarding x and y , E denotes the function of mechanical properties regarding x and y . Both K and E are calculated by the surrogate model as introduced in Section 3.2.

3.3.2. The workflow for LLM's intervention

Fig. 4 illustrates the workflow of the LLM-assisted optimization framework. In each generation, an offspring population is first generated using standard genetic operators. Every fixed number of generations (denoted as F), the LLM module is activated to intervene in the evolutionary process.

The LLM intervention consists of several steps:

- Prompting the LLM with elite solutions:** The top 10 elite solutions from NSGA-II are selected from the current Pareto front. These elite individuals, along with their corresponding objective values, are used to construct a prompt that is passed to the LLM.
- Inference and solution generation:** Using the constructed prompt (details in Section 3.3.3), the LLM is called to reason upon this context and produces a set of new candidate solutions (denoted as N) in the form of seed coordinates. These solutions are text-based outputs representing 10-dimensional decision vectors.
- Injection:** The LLM-generated outputs are parsed and validated. Valid candidate solutions are then injected into the offspring population by replacing the worst-performing individuals previously generated by traditional genetic operators. This hybrid offspring population is then subject to environmental selection.

The search continues iteratively, with each generation producing offsprings via crossover and mutation, being assisted periodically with LLM-generated solutions. The optimization progress terminated when 100 generations are reached.

In this study, we selected ChatGPT-4o (Overview - OpenAI API, n.d.) as the LLM for generating candidate solutions. The model was accessed via the official OpenAI API endpoints, which provide a stable and scalable interface for LLM integration.

3.3.3. Structured knowledge representation

In this section, we present the detailed knowledge representation strategy given to LLM. An example of the final prompt template used in

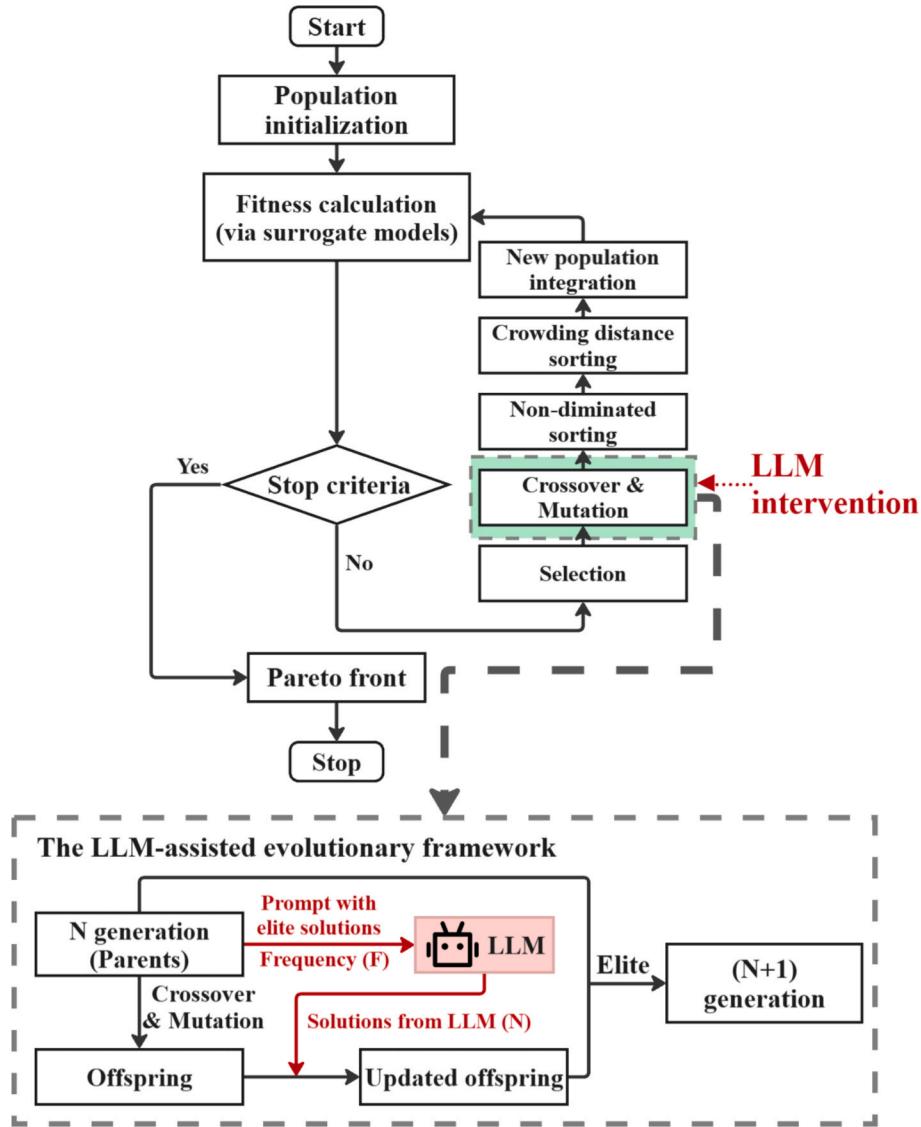


Fig. 4. Workflow for the LLM-assisted framework in this study.

our framework is shown in Fig. 5. The prompt is engineered to mimic the decision-making context of a structural expert. The key components are:

- (i) *Task framing and design requirements*, where the prompts clearly frame the tasks for the LLM, describe the decision variable format, and outline the design goals;
- (ii) *In-context examples*, which show a few top-performing solutions from the current population containing the seed configuration and its corresponding objective values;
- (iii) *Output instructions*, with the instructions for LLM to generate the next generation in a specific format. These rules ensure that candidates generated by the LLM are usable in surrogate-based evaluations without additional post-processing.

In this prompt, knowledge representation is realized through a structured prompt that encodes three levels of expert information: (i) Domain constraints, defining the geometric and manufacturability boundaries; (ii) Performance objectives, representing the physical goals; and (iii) Historical context, where elite individuals from the current population serve as dynamic knowledge exemplars for the reasoning process.

3.4. Experimental setup

This section delineates the experimental settings of compared multi-objective algorithms and the associated performance metrics. To investigate the impact of LLM intervention strategies, we conduct nine distinct tests by varying key hyperparameters: three levels for the number of elite solutions provided to the LLM ($N = 5, 10, 20$) and three for the LLM intervention frequency (every $F = 5, 10$, or 20 generations). The performance of the proposed LLM-assisted framework is compared against the baseline NSGA-II algorithm to assess the benefits of incorporating LLM-driven solution generation. The detailed parameter configurations for all algorithms are summarized in Table 3. To ensure statistical robustness, each configuration was executed over 10 independent runs under identical settings.

To evaluate the efficacy of the proposed framework, two widely adopted multi-objective optimization indicators are employed: Hypervolume (HV) (Bradstreet, 2011) and Inverted Generational Distance (IGD) (Sun et al., 2019), with their definitions provided in Table 4.

4. Results and discussions

In this chapter, we showcase the comparative performance analysis

Knowledge representation for LLM

(i) Task framing and design requirements

You are a creative design engineer specializing in material science. Your task is to generate new sets of seed points for a Voronoi diagram to define a hollow concrete block's macrostructure.

Physical context to guide your design:

1. The 5 (x, y) seed points define a Voronoi diagram.
2. The edges of this diagram become the solid concrete frame.
3. The cells of this diagram become the air cavities for insulation.

The goal is to find the coordinates of 5 seed points that balance thermal insulation (minimize conductivity) with mechanical strength (maximize Young's modulus)

(ii) In-context examples

Here are the best-performing solutions found so far. Analyze the patterns in the seed locations and their resulting objective values.

Solution {i}:

Seeds: (x1,y1,x2,y2,x3,y3,x4,y4,x5,y5)

Objectives: Thermal Conductivity = ..., Young's Modulus = ...

(iii) Output instructions

Based on your expert analysis of the provided solutions and the underlying physics, generate EXACTLY N new, diverse, and high-potential solutions.

Your goal is to intelligently explore the search space to find solutions that could significantly improve the Pareto front.

Follow these constraints STRICTLY:

1. Coordinate bounds: All x and y coordinates must be within the [0.0, 50.0] range.
2. Manufacturability: The minimum pairwise distance between any two seed points in a single solution must be at least 10.0.
3. Data type: All coordinate values MUST be floating-point numbers with decimal precision (e.g., 2 decimal places).
4. Output format: Your entire output must be a single, valid JSON list of lists. Do not include any other text, explanations, or markdown formatting. Each inner list represents one solution and must be a flat list of 10 numbers representing 5 (x, y) coordinate pairs, in the format [x1, y1, x2, y2, x3, y3, x4, y4, x5, y5].
5. Ensure that the seeds are not aligned (i.e., not colinear), and are well distributed across the domain to form valid Voronoi cells.

Output format: Each solution should be a list of 10 numbers.

Format: [[x1,y1,x2,y2,x3,y3,x4,y4,x5,y5], [next solution], ...]

Example of a single solution's format:

[0.02, 31.62, 13.49, 25.61, 39.18, 48.31, 49.82, 35.10, 4.99, 0.45]

Only output the solutions in the specified format, no explanations.

Fig. 5. An example of prompts for ChatGPT-4o during the optimization.

Table 3

Parameters of algorithms in this paper.

	Category	Value setting
Parameters for all algorithms	Number of generations	100
	Population size	50
	Crossover rate	0.9
	Mutation rate	0.1
Parameters specific to the LLM-assisted framework	LLM employed	ChatGPT-4o
	Number of elite solutions given to LLM	10
	Temperature setting for the LLM	0.5
	Number of LLM intervention frequency (F)	5/10/20
	Number of solutions generated by LLM (N)	5/10/20

Table 4

Performance metrics for evaluation.

Metric	Description
Hypervolume (HV)	It measures the volume in the objective space dominated by the obtained Pareto front relative to a reference point. A higher HV implies a more favorable outcome.
Inverted Generational Distance (IGD)	It evaluates the average Euclidean distance from a set of true Pareto-optimal points to the solutions obtained by the algorithm. A lower IGD value indicates a higher-quality solution set.

of the compared algorithms to assess the efficacy of our proposed LLM-assisted framework within the domain of optimization for hollow concrete components.

4.1. Overall performance comparison and attained pareto fronts

This section shows the overall performance comparison of the algorithms. We first provide a qualitative assessment of the optimization performance through a direct visual inspection of the final Pareto fronts. Fig. 6 illustrates the non-dominated solution sets obtained from 10 independent runs for both the standard NSGA-II (black points) and the proposed LLM-assisted framework (red points) across various LLM intervention parameters. The ideal region in the objective space is the top-left corner, representing simultaneously minimized objective 1 (thermal conductivity) and maximized objective 2 (Young's modulus). As evident from Fig. 6, the Pareto fronts generated by the LLM-assisted framework consistently exhibit a visible shift towards the ideal top-left region of the objective space. This indicates superior performance, as for nearly any solution identified by the standard NSGA-II, a dominating solution can be found on the front achieved by our proposed framework.

A particularly significant finding can be observed at the low-conductivity end of the front (i.e., towards the left extreme of objective 1). The standard NSGA-II's exploration appears to halt at a thermal conductivity (objective 1) of approximately 0.50. In contrast, the LLM-assisted framework successfully discovers a number of sets of elite solutions with objective 1 values lower than 0.50. This breakthrough is especially pronounced in cases where the number of LLM-generated solutions (N) is higher or the intervention frequency (F) is lower. This observation strongly suggests that when the standard algorithm

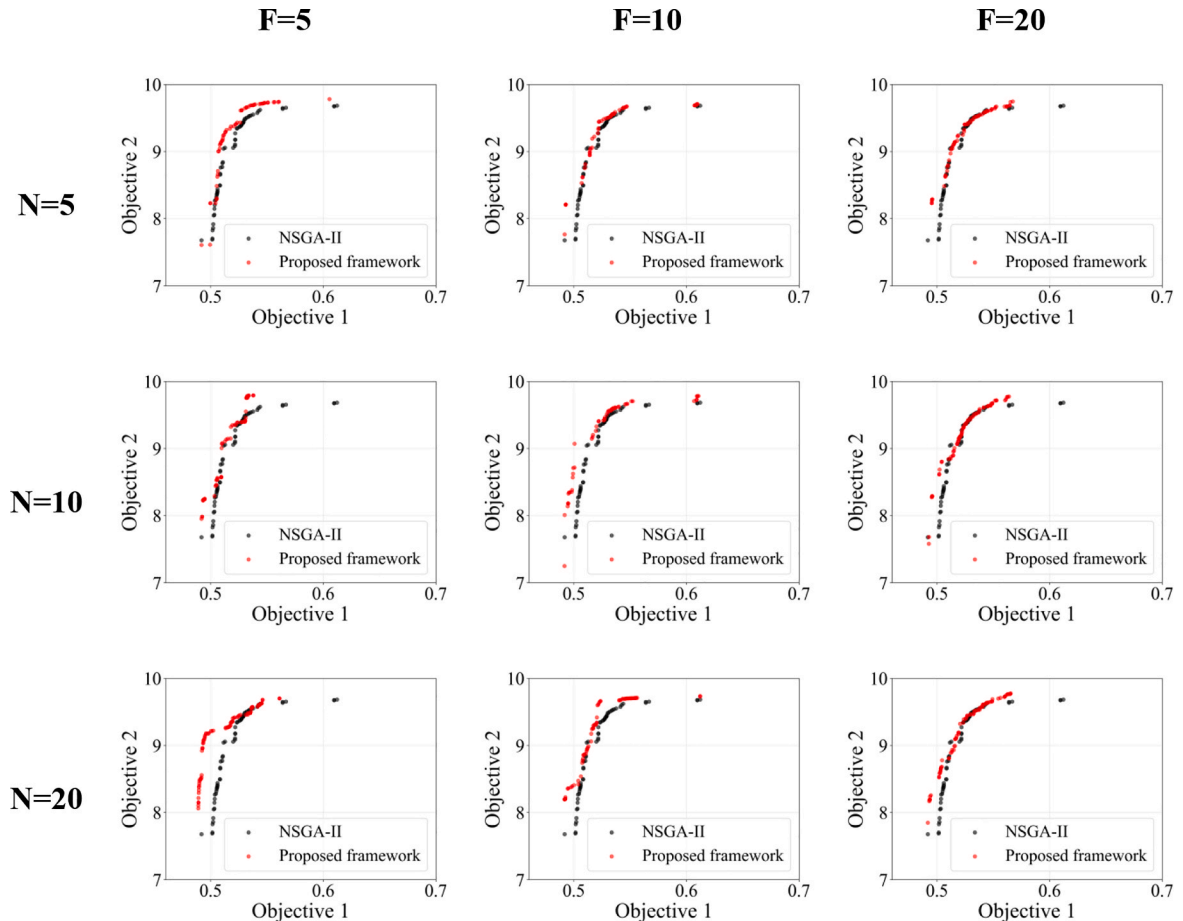


Fig. 6. Comparison of the final Pareto fronts from 10 independent runs of the standard NSGA-II (black) and the LLM-assisted framework (red). The ideal objective is to reach the top-left corner (low thermal conductivity, high Young's modulus).

struggles to identify the specific topological features required to overcome a certain performance barrier, the candidate solutions introduced by the LLM are instrumental in guiding the search out of such a trapped basin and into a new, superior region of the design space.

Furthermore, in the central region of the Pareto front, which represents the most balanced trade-offs between the two conflicting objectives, the performance of the LLM-assisted framework shows nuanced behavior depending on the intervention frequency. Specifically, when the LLM intervention frequency is lower ($F = 20$), the LLM-assisted framework's performance in this central region is largely comparable to that of the standard NSGA-II. However, in cases with higher LLM intervention frequencies, the LLM-assisted framework outperforms the traditional algorithm more demonstrably in identifying better trade-off solutions. This provides compelling evidence that the LLM-assisted framework is capable of identifying effective trade-off solutions for multi-objective problems with appropriate intervention strategies.

To further evaluate the effectiveness of the proposed LLM-assisted optimization framework, we then examine its performance on the IGD, which quantifies how closely the obtained solutions approximate the true Pareto front. As shown in Table 5, the baseline NSGA-II yields an average IGD of 0.1254 with a standard deviation of 0.0505. In contrast, all configurations of the LLM-assisted framework produce lower IGD values, with the best-performing setting achieving an average of 0.0866 with a standard deviation of 0.0141. This corresponds to a substantial 30% reduction in IGD, suggesting that the LLM-generated solutions effectively nudge the evolutionary search toward regions closer to the true Pareto front.

Table 6 presents the HV values achieved by NSGA-II and the LLM-assisted framework over 10 independent runs. The HV results strongly reinforce the conclusions drawn from the IGD data and provide additional insights: the LLM-assisted framework consistently achieves higher HV scores across all tested configurations compared to the baseline algorithm, with the best reaching 0.6584 (Std = 0.0299). This represents a consistent improvement in the dominated objective space, which again confirms the better overall quality of the solution sets found by the LLM-assisted framework.

4.2. Statistical validation of performance differences

In this section, statistical analysis is performed using the Friedman test and the Wilcoxon test (Chen et al., 2025) to evaluate the performance among 10 algorithm configurations.

(i) Friedman test

The Friedman test was conducted to investigate the presence of significant differences among the algorithms. Results show that the Friedman statistic $\chi^2 = 24.26$, with a corresponding p-value of 0.003908 < 0.05. It shows that there are significant performance differences between at least two of the compared algorithms.

(ii) Wilcoxon test

Pairwise comparisons between NSGA-II and each LLM configuration were conducted using the Wilcoxon test. The resulting p-values are

Table 5

Comparison of IGD values for standard NSGA-II and the LLM-assisted framework over 10 independent runs.

Type	F	N	Average	Std
NSGA-II	0	0	0.1254	0.0505
LLM-assisted framework	5	5	0.0927	0.0262
		10	0.1054	0.0369
		20	0.1124	0.0269
	10	5	0.1024	0.0275
		10	0.0866	0.0141
		20	0.1029	0.0404
	20	5	0.1168	0.0520
		10	0.1148	0.0434
		20	0.0998	0.0401

Table 6

Comparison of HV values for standard NSGA-II and the LLM-assisted framework over 10 independent runs.

Type	F	N	Average	Std
NSGA-II	0	0	0.6135	0.0407
LLM-assisted framework	5	5	0.6437	0.0290
		10	0.6530	0.0208
		20	0.6530	0.0456
	10	5	0.6347	0.0232
		10	0.6584	0.0299
		20	0.6414	0.0330
	20	5	0.6179	0.0360
		10	0.6273	0.0321
		20	0.6264	0.0389

summarized in Table 7, where a smaller p-value denotes a more significant difference. Results indicate that the configurations LLM-assisted ($F = 5$, $N = 10$) and LLM-assisted ($F = 10$, $N = 10$) demonstrated a statistically significant improvement over the standard NSGA-II, with p-values of 0.0195 and 0.0020, respectively. This confirms that under these specific intervention strategies, the integration of LLM-generated solutions robustly enhances the optimization performance.

Several other configurations, including LLM-assisted ($F = 5$, $N = 5$) ($p = 0.0840$), LLM-assisted ($F = 5$, $N = 20$) ($p = 0.0645$), and LLM-assisted ($F = 10$, $N = 20$) ($p = 0.0840$), yielded p-values below the 0.1 level, indicating marginal statistical significance. While these results do not cross the conventional 0.05 threshold, they nonetheless suggest that these configurations may offer practical benefits, especially when considered alongside the improved average HV and IGD values. In contrast, configurations with the lowest intervention frequency ($F = 20$) showed less consistent advantages, with none achieving statistical or marginal significance. This suggests that a too conservative intervention schedule may underutilize the generative capacity of the LLM, failing to provide substantial guidance to the optimization process.

In summary, while not all LLM-assisted configurations led to statistically significant improvements, the overall pattern of superior performance reinforces the value of incorporating LLMs into evolutionary optimization.

4.3. Convergence behavior analysis

Complementing the tabulated results, this section visually charts the convergence paths of the average HV evolution across 10 independent runs for the standard NSGA-II and the LLM-assisted framework. As shown in Fig. 7, the vertically dashed blue lines indicate the generations where LLM were activated to intervene the evolution. It can be clearly observed that the LLM-assisted framework gradually converges to a higher HV than the standard NSGA-II as the evolution process progresses. This further demonstrates that the injection of LLM's solutions enables a much more efficient exploration of the search space.

A closer inspection of the convergence curves reveals a nuanced

Table 7

Results of Wilcoxon signed-rank test.

Baseline algorithm: NSGA-II		
Compared algorithms	p-value	Significance
LLM-assisted ($F = 5$, $N = 5$)	0.0840	*
LLM-assisted ($F = 5$, $N = 10$)	0.0195	**
LLM-assisted ($F = 5$, $N = 20$)	0.0645	*
LLM-assisted ($F = 10$, $N = 5$)	0.1934	
LLM-assisted ($F = 10$, $N = 10$)	0.0020	***
LLM-assisted ($F = 10$, $N = 20$)	0.0840	*
LLM-assisted ($F = 20$, $N = 5$)	0.6953	
LLM-assisted ($F = 20$, $N = 10$)	0.2324	
LLM-assisted ($F = 20$, $N = 20$)	0.2754	

Notes: *: statistically significant at the 0.1 level; **: statistically significant at the 0.05 level; ***: statistically significant at the 0.01 level.

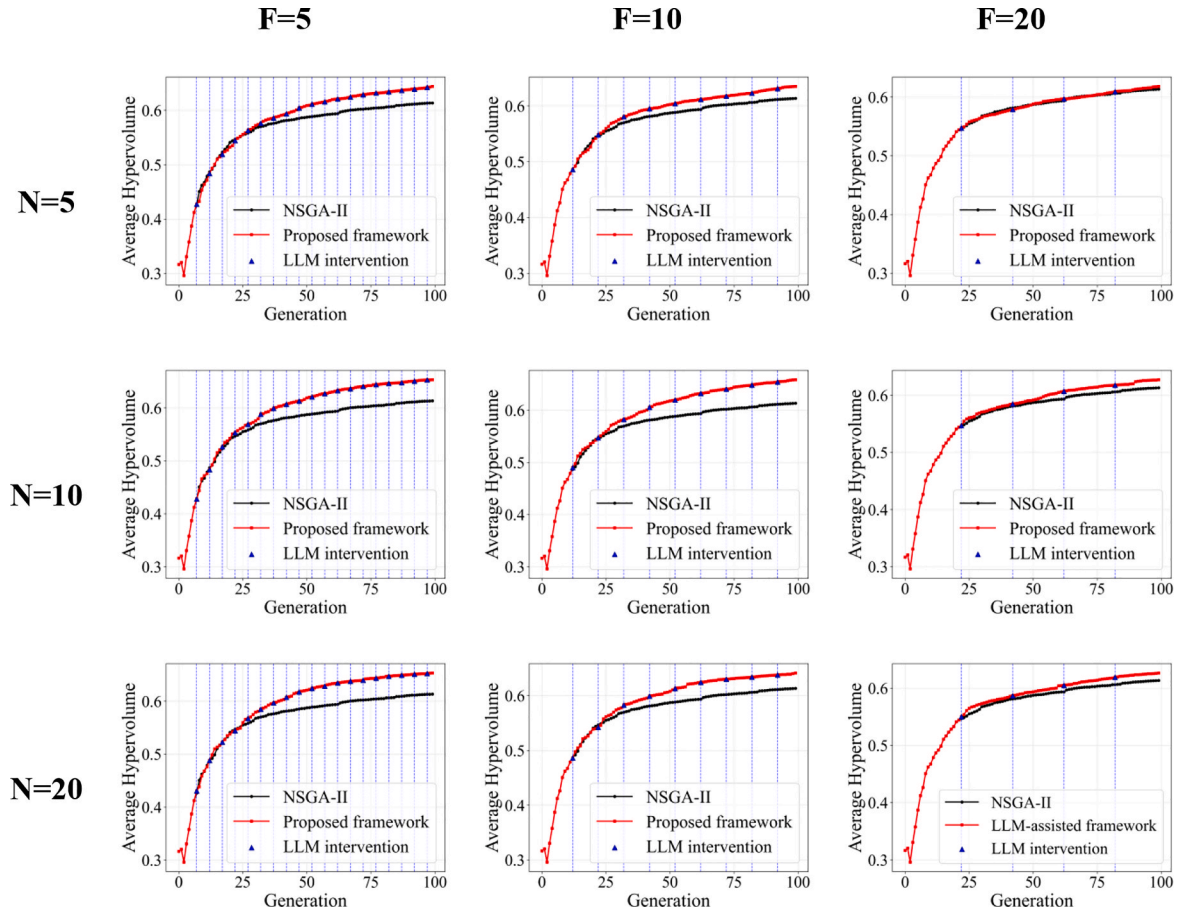


Fig. 7. Comparison of average HV evolution from 10 independent runs of the standard NSGA-II (black) and the LLM-assisted framework (red).

dynamic regarding the immediate impact of LLM solution injection. While the cumulative benefit of LLM intervention is clear, the HV does not always exhibit a dramatically abrupt increase precisely at the first intervention. Instead, the LLM-assisted framework often demonstrates a sustained and accelerated improvement in HV over subsequent generations following an injection, rather than an instantaneous large jump. This phenomenon can be attributed to several factors: The solutions introduced by the LLM, while potentially high-quality, require a few subsequent generations to propagate their full impact throughout the population via traditional genetic operators. These injected solutions introduce beneficial genes and guide the search towards previously unexplored, promising regions of the design space. They help prevent premature convergence and provide the necessary diversity to break out of local optima that the standard NSGA-II might otherwise get trapped in. Therefore, the full manifestation of these benefits is often a cumulative effect that unfolds over time as the population integrates these LLM-derived insights.

4.4. Supplementary analysis

This section presents three supplementary analyses to provide a more comprehensive characterization of the proposed LLM-assisted framework. Section 4.4.1 examines the behavioral contribution of the LLM to the optimization process through a quantitative analysis of solution-level diversity in the design space, offering empirical evidence for the LLM's role as a structured exploration mechanism. Section 4.4.2 investigates the long-term impact of the LLM replacement mechanism on population diversity, addressing the concern that periodic replacement of worst-performing individuals may induce premature convergence. Section 4.4.3 benchmarks the performance of the proposed framework

against both traditional multi-objective optimization algorithms and representative LLM-assisted evolutionary approaches reported in the current literature, contextualizing the results achieved within the broader state of the art.

4.4.1. Behavioral characterization of LLM-assisted exploration

While the internal reasoning process of the LLM remains implicit by nature, its behavioral contribution to the optimization process can be characterized through quantitative analysis of the design space. To this end, we examined the relative position of LLM-generated solutions with respect to the elite set in the Voronoi seed design space and related this to the observed optimization trajectory.

For each LLM intervention event, we computed the mean Euclidean distance (Mesquita et al., 2017) from the LLM-generated solutions to their nearest elite solution, and compared this to the corresponding mean distance from the GA offspring to the same elite set. A larger distance indicates that a group of solutions occupies regions of the design space that are more different from the current high-performing solutions, reflecting broader exploratory coverage. This analysis was conducted across 10 independent runs, and statistical significance was assessed using a paired *t*-test.

The results are summarized in Table 8. Across the 10 independent runs, LLM-generated solutions are consistently more different from the elite set than GA offspring, indicating that the LLM systematically generates candidates in underexplored regions of the design space rather than locally perturbing known high-performing solutions. This difference is statistically significant at the 0.01 level in 8 out of 10 runs, and at the 0.05 or 0.1 level in a further 2 runs. The one exception (Run 3) falls short of conventional thresholds but does not contradict the overall trend.

Table 8

Generation-level paired *t*-test results: LLM-generated solutions vs. GA offspring, distance to elite set.

Run	Mean distance to elite Δ (LLM – GA)	p-value	Significance
1	47.0475	1.0643E-06	***
2	33.2570	4.1428E-03	***
3	12.6305	1.3598E-01	
4	34.0473	3.3999E-05	***
5	51.2308	1.7695E-08	***
6	36.8491	4.8212E-06	***
7	17.2611	5.9712E-02	*
8	53.8134	7.1376E-08	***
9	14.2131	1.4568E-02	**
10	38.9667	1.9087E-03	***

Notes: Δ (LLM – GA) denotes the difference in mean Euclidean distance to the current elite set between LLM-generated solutions and GA offspring; a positive value indicates that LLM-generated solutions are more different from the elite set, reflecting broader coverage of underexplored regions in the design space. Statistical significance is assessed via paired *t*-test across LLM intervention events within each run. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

These results indicate that the LLM does not replicate or locally perturb known high-performing designs. Instead, it consistently generates candidates in previously underexplored regions of the design space. This behavioral pattern suggests that the LLM functions as a structured exploration mechanism, complementing the inherent tendency of GA crossover and mutation operators. Such a behavioral pattern is interpreted as a heuristic reasoning mechanism: by analyzing the structural patterns of elite Voronoi seed configurations encoded in the prompt, the LLM reasons over spatial relationships in the design space and proposes geometrically diverse candidates that differ systematically from the current elite solutions. This process is operationalized through in-context learning (Dong et al., 2022), in which the elite solutions serve as implicit demonstrations that guide the LLM's generation, simulating the pattern-based inference of a domain expert drawing on accumulated experience, rather than stochastic random search. In this sense, the LLM's contribution is not random diversification, but knowledge-informed exploration grounded in an implicit understanding of the design space. This observation also aligns with the conclusion in Ref (Guo et al., 2026), that LLM facilitates more proactive exploration when the solution space is ample and diverse.

This behavioral pattern is also consistent with the optimization trajectory observed in Fig. 7. As shown, the HV of the LLM-assisted framework begins to diverge favorably from that of standard NSGA-II from the generation of the first LLM intervention onward, and this advantage persists and accumulates throughout the remaining optimization process. The temporal alignment between LLM intervention events and the onset of sustained performance improvement supports the interpretation that the LLM's exploratory behavior has a lasting positive influence on the evolutionary trajectory, as the diverse candidates introduce broader the population's coverage of the Pareto front and provide raw material from which subsequent GA operators can further refine high-quality solutions.

Taken together, the solution-level distance analysis and the HV trajectory provide complementary behavioral evidence that the LLM contributes to superior Pareto fronts by introducing structured exploration into the search process, covering regions of the design space that GA operators alone would not readily access.

4.4.2. Impact of LLM's intervention on population diversity

A potential concern with the LLM replacement mechanism is that it may reduce population diversity over time, potentially leading to premature convergence and weakened global search capability. To examine this, we conducted a long-term diversity analysis comparing the population diversity of the LLM-assisted framework ($F = 10$, $N = 10$) against that of standard NSGA-II across 10 paired runs.

Population diversity was quantified using the mean pairwise Euclidean distance (Mesquita et al., 2017) among the 50 individuals in the population at each monitored generation, corresponding to the LLM intervention points. For each generation, a Welch's *t*-test was performed to assess whether the diversity of the population from the LLM-assisted framework differed significantly from that of the standard NSGA-II's population.

The results are summarized in Table 9. At all monitored generations, no statistically significant difference in population diversity was found between the two frameworks (all $p > 0.05$; overall mean p -value = 0.693). Notably, the mean diversity of the LLM-assisted framework's population is numerically higher than that of standard NSGA-II at the majority of monitored generations, and the observed differences remain small in magnitude throughout the 100-generation optimization process. These findings indicate that the LLM replacement mechanism does not introduce systematic diversity loss over the course of optimization, and the concern of premature convergence induced by LLM intervention is not supported by the empirical evidence.

4.4.3. Comparison with the literature

To assess the effectiveness of the proposed LLM-assisted evolutionary framework, its performance was benchmarked against both traditional multi-objective optimization algorithms and existing LLM-assisted evolutionary approaches. HV was adopted as the primary comparison metric, as it simultaneously captures both the convergence and diversity of the obtained Pareto front. All experiments were conducted under identical settings: population size of 50, 100 generations, and 10 independent runs. Table 10 summarizes the mean and standard deviation of HV values across all compared methods.

Among the traditional baselines, NSGA-II achieves a mean HV of 0.6135, demonstrating solid optimization capability on this problem. MOEA/D, despite being a widely adopted decomposition-based algorithm, yields a substantially lower mean HV of 0.1447 with a high standard deviation of 0.0989. It should be noted that MOEA/D does not natively support constrained optimization; constraint handling was therefore incorporated via a penalty function formulation. This modification, combined with MOEA/D's fixed weight-vector decomposition strategy, likely constrains its ability to maintain sufficient population diversity in this high-dimensional, geometrically constrained design space, which may partially account for its substantially lower HV performance.

Among the LLM-assisted approaches, two alternative integration strategies were implemented as baselines: (1) using the LLM to replace genetic crossover and mutation operators throughout the optimization process, and (2) using the LLM solely to generate the initial population, after which standard NSGA-II takes over. The former achieves a mean HV of 0.6270, exceeding NSGA-II by 2.2%, while the latter yields a slightly lower mean HV of 0.6081, with notably reduced variance ($\text{Std} = 0.0197$).

The framework proposed in this paper achieves the highest mean HV

Table 9

Per-generation diversity comparison between LLM-assisted GA and standard NSGA-II populations (mean pairwise Euclidean distance, $n = 10$ runs per group).

Generation	NSGA-II mean diversity	LLM-assisted framework mean diversity	p-value
11	43.3092	43.9331	0.9132
21	35.8571	39.0447	0.4161
31	32.6158	34.0828	0.7416
41	31.5470	28.4173	0.5195
51	30.9162	31.4214	0.9086
61	30.7719	31.5203	0.8709
71	28.8872	32.0792	0.4873
81	27.5109	29.3033	0.7231
91	27.0511	27.4953	0.9320

Note: Welch's *t*-test; no significant difference at any generation (all $p > 0.05$).

Table 10
Comparison of HV performance across optimization frameworks.

Category	Description	Mean HV	Std HV	References
Traditional methods	NSGA-II	0.6135	0.0407	Deb et al. (Deb et al., 2002)
	MOEA/D	0.1447	0.0989	Huang and Li (Zhang and Li, 2008)
LLM-assisted methods	LLM for genetic operators' replacement	0.6270	0.0420	Meyerson et al. (Meyerson et al., 2024), Brahmachary et al. (Brahmachary et al., 2024)
	LLM for population initialization	0.6081	0.0197	Huang et al. (Huang et al., 2025)
	Framework in this paper	0.6584	0.0299	—

of 0.6584, representing a 7.3% improvement over the best-performing LLM-assisted baseline (LLM for genetic operators' replacement). Furthermore, the proposed framework maintains a competitive standard deviation of 0.0299, indicating stable and consistent performance across independent runs. These results collectively demonstrate that the periodic LLM intervention strategy adopted in this paper achieves a superior balance between exploration and exploitation compared to both full replacement and initialization-only strategies. The ability of the LLM to introduce structured exploration into the evolutionary search, as evidenced by the behavioral analysis in Section 4.4.1, is a key contributor to this performance advantage.

4.5. Insights on practical applications

In this work, we assessed two critical parameters governing LLM intervention: the intervention frequency (F) and the number of generated solutions (N). This section analyzes their impact on optimization performance and discusses the practical trade-offs involved in real-world applications.

(i) LLM intervention frequency (F)

The experimental results reveal a nonlinear relationship between intervention frequency and optimization performance. As evidenced by the HV and IGD metrics (Table 5 and 6), all tested frequencies ($F = 5, 10, 20$) result in a performance improvement over the baseline NSGA-II. More importantly, a general trend indicates that more frequent interventions (i.e. lower F values) generally lead to better performance. However, this improvement in performance must be weighed against the associated computational cost, which is primarily driven by API calls for LLM and token consumption. Moving from $F = 10$ to $F = 5$ doubles the frequency of LLM interventions, thus approximately doubling the LLM calls and token usage. Results show that though the performance of $F = 5$ does improve, the gain is marginal and far from double. This demonstrates a classic case of diminishing returns, that the intervention frequency of LLM is not linear with the associated improvement in the final solution quality. Additionally, this pattern suggests that there may exist an optimal intervention frequency beyond which additional LLM calls provide increasingly limited benefits. The diminishing returns can be attributed to several reasons: (i) The evolutionary algorithm requires sufficient generations to effectively integrate and exploit LLM-generated solutions. Excessive intervention frequency may disrupt this natural adaptation process; (ii) As the population converges toward optimal regions, frequent LLM interventions may generate increasingly similar solutions and reduce their contribution.

For practical engineering applications, this leads to a crucial consideration on the cost-benefit trade-off. The choice of intervention frequency should be driven by the specific constraints and requirements of the application. In terms of the case in this work, the $F = 10$ configuration emerges as a highly efficient choice, offering a near-optimal

performance boost at a moderate computational cost. While $F = 5$ can provide a slight additional edge, it may only be justified in scenarios where achieving the absolute maximum performance is critical and cost is not a primary constraint. Importantly, all configurations provide substantial value over traditional NSGA-II, offering practitioners sufficient robustness as well as flexibility in parameter selection.

(ii) Number of solutions generated by the LLM (N)

The results of the number of solutions generated by the LLM (N) demonstrate the framework's robustness. Across all tested values ($N = 5, 10, 20$), the framework consistently outperforms the baseline NSGA-II. While performance variations do exist across different N values, these differences do not compromise the framework's overall effectiveness. $N = 5$ exhibits slightly inferior performance compared to those with higher N values, which suggests that a lower level of intervention may be insufficient to fully leverage the LLM's generative potential.

The relative insensitivity to the N parameter represents a significant practical advantage for deployment. The framework's consistent effectiveness indicates that practitioners can implement the method without extensive parameter exploration, significantly reducing the setup time and computational overhead typically associated with method deployment. The robustness across N values ensures that even suboptimal parameter choices still deliver performance gains.

In practical applications, it is important to note that token usage of the LLM scales positively with the number of generated solutions. This direct relationship between N and computational cost provides practitioners with a clear mechanism for balancing performance against resource constraints. The adjustment of N can therefore be based on immediate computational budgets or problem-specific requirements while maintaining confidence in the framework's effectiveness.

4.6. Summary of findings

From a knowledge systems perspective, the proposed framework can be understood as a knowledge-intensive expert system for multi-objective structural optimization. The prompt engineering mechanism constructs a structured knowledge representation that communicates geometric constraints and optimization history to the LLM at each intervention. The LLM then applies a heuristic reasoning mechanism to generate candidate solutions that extend the search into underexplored regions of the design space. This architecture mirrors the knowledge acquisition, representation, and reasoning pipeline of classical expert systems (Shang and Systems, 2005), while extending it to the domain of multi-objective evolutionary optimization.

The results presented in this section provide compelling evidence for the efficacy and superiority of the proposed LLM-assisted framework. Qualitatively, a visual inspection of the Pareto fronts revealed a consistent shift towards the ideal objective space, with the framework demonstrating a unique ability to break through performance barriers where the traditional NSGA-II stagnated. This qualitative observation was robustly supported by a comprehensive statistical analysis, which confirmed that the LLM-assisted framework achieved a statistically significant improvement in both HV and IGD under specific intervention strategies. Furthermore, our analysis of the convergence paths highlighted that the LLM's contribution is not merely an instantaneous boost but a cumulative effect that intelligently guides the evolutionary process over generations. These findings collectively demonstrate that by leveraging the intelligent contributions of LLMs, it is possible to achieve higher-quality solutions and a more efficient and effective optimization of complex geometric designs. This highlights a novel pathway for achieving superior performance not solely through traditional adjustments of genetic operators, but through a new and more intuitive approach. The adjustment of LLM's output, being controlled by natural language instructions, offers a potentially more user-friendly paradigm for optimizing complex designs compared to the opaque tuning of traditional genetic algorithm hyperparameters, thereby offering a powerful new alternative for knowledge-intensive engineering tasks.

5. Conclusive remarks

5.1. Summary

This study proposed and validated a novel knowledge-intensive, LLM-assisted evolutionary framework for the multi-objective geometric optimization of high-performance hollow concrete blocks characterized by irregular polygonal air cavities. By leveraging the advanced reasoning and generative capabilities of an LLM to periodically introduce new candidate solutions into the NSGA-II algorithm, the proposed framework demonstrated a significant performance enhancement over the traditional baseline. The results, supported by rigorous statistical analysis, confirmed that the LLM-assisted approach was able to produce superior Pareto fronts and achieved a statistically significant improvement in optimization metrics. A key contribution of this work lies in its specific focus on the AEC domain, where tasks often involve highly knowledge-intensive reasoning around spatial relationships, structural feasibility, and performance trade-offs. The proposed framework demonstrates that LLMs can effectively support such cognitive design processes by generating geometrically viable and high-performing structural configurations, thus bridging human-level reasoning with computational search.

Furthermore, we conducted a systematic comparative analysis among cases with two key parameters related to LLM: intervention frequency F and the number of generated solutions N . The results showed that the efficacy of the LLM in augmenting the evolutionary search is dependent on these parameters, with specific configurations (e.g., $F = 10$, $N = 10$) yielding the most robust and statistically significant gains. This finding provides crucial, data-driven guidance for practitioners seeking to apply similar hybrid optimization methods. This research establishes a new paradigm for supporting knowledge-intensive engineering design tasks, where optimization is no longer solely reliant on fine-tuning traditional genetic operators. Instead, the integration of natural language-driven reasoning enables a more intuitive and efficient search process, thereby lowering the barrier to high-quality design exploration in complex geometric contexts.

5.2. Limitations and future work

While this study demonstrates the efficacy of the LLM-assisted framework in enhancing multi-objective evolutionary optimization for hollow concrete component design, several limitations remain and point to productive directions for future research.

First, the transparency of the LLM component. A key limitation of the proposed framework concerns the mechanistic interpretability of the LLM's generative behavior. While the behavioral analysis in [Section 4.4.1](#) demonstrates that LLM-generated solutions consistently occupy underexplored regions of the design space, the internal reasoning process by which the LLM interprets the encoded Voronoi seed configurations, including how it infers geometric relationships or avoids infeasible arrangements, remains implicit and cannot be directly inspected. This opacity is an inherent property of LLMs in their current form and is not specific to the present application ([Huang et al., 2024](#)). Future work will investigate interpretability mechanisms, including structured prompt analysis, attention visualization, and comparative ablation across prompt designs, to provide more granular insight into which aspects of the encoded design context most strongly influence the LLM's generative behavior. This line of inquiry is expected to strengthen the alignment of LLM-assisted optimization frameworks with the explainability requirements of knowledge-based expert systems.

Second, prompt engineering and LLM configuration. The exploration of prompt engineering strategies was not a primary focus of this study, yet it is a critical component of human-LLM interaction. Future work should explore advanced prompt engineering techniques, such as chain-of-thought reasoning and multi-agent architectures ([Han et al., 2025](#)), to investigate their impact on solution quality and interpretability.

Additionally, this study did not systematically investigate different LLM types or inference temperatures. Future research could examine LLMs capable of producing explicit reasoning traces, which would further improve the transparency of the generative process. An adaptive temperature mechanism that dynamically adjusts the LLM's exploratory tendency based on the real-time state of the population also presents a promising direction for better balancing exploration and exploitation.

Third, surrogate model approximation. The use of GNN surrogate models introduces an inherent approximation error relative to high-fidelity FEA. While the validated prediction accuracy (MRE: 1.26% and 4.80% for thermal conductivity and Young's modulus, respectively) demonstrates that this error is well within acceptable bounds for the present application ([Granacher et al., 2021](#); [An et al., 2015](#)), surrogate approximation may in principle influence the objective values perceived by the LLM during solution generation. It should be noted that all candidate solutions are independently re-evaluated by the surrogate prior to population entry, ensuring that approximation errors are not accumulated across generations. Nevertheless, the integration of uncertainty quantification methods or adaptive surrogate updating strategies to further mitigate this effect represents a promising direction for future research.

Fourth, domain knowledge integration. The current approach can be considered a form of zero-shot learning for the LLM: while it successfully leverages the model's pre-trained understanding of spatial relationships, integrating pre-trained LLMs with more domain-specific knowledge ([Jiang et al., 2024](#)) may further enhance the coupling between LLMs and EAs, particularly for structural optimization problems where material science and engineering constraints are complex and highly specialized.

CRedit authorship contribution statement

Zhuyin Lu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Hanmo Wang:** Data curation, Formal analysis, Investigation, Methodology, Validation. **Jingwen Song:** Investigation, Methodology, Validation, Visualization. **Xin Tian:** Investigation, Methodology, Validation. **Alexander Lin:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alexander Lin reports financial support was provided by Government of Singapore Ministry of Education. Alexander Lin reports a relationship with Government of Singapore Ministry of Education that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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